

Activity 9 Pre-Lab — EAS Bromination of Aniline / N-Phenylacetamide

[Your name] & [partner]
[MM/DD/YY]

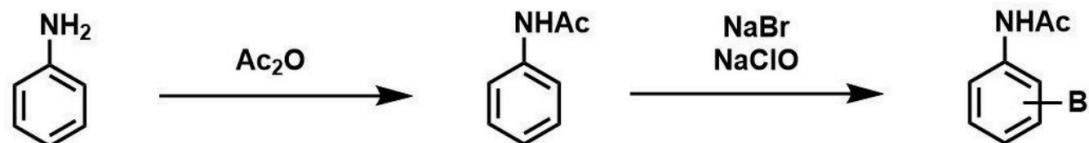
Organic Lab · electrophilic aromatic substitution

■ Black = write this in your notebook ■ Blue = context, don't copy it Copy the header onto every page.

Purpose

Study **electrophilic aromatic substitution**: first **protect** aniline's amine as an amide (N-phenylacetamide), then **brominate** the ring. Shows that -NHR is a strong **o/p-activator** and that protecting it tempers the reactivity enough to get clean mono-**para** product.

🔪 DRAW — the two steps (from the procedure)



Step 1 protects the amine as the amide (Ac_2O); Step 2 brominates — Br^+ made in situ from $\text{NaBr} + \text{NaOCl}$ (bleach) in acetic acid; the amide is o/p-directing → para major.

Chemical Reagents & Safety Table

CONTEXT ▶ Required (8). Typical SDS values — cross-check the SDS tile. **Aniline is toxic & absorbs through skin.**

Chemical	Role	MW	Amount	Physical properties	GHS pictograms
Aniline	Starting material (limiting)	93.13	~1 mL (0.5 mL x2)	Colorless–brown oily liquid; bp 184 °C; d 1.02 g/mL	
Acetic anhydride	Acetylating (protecting) agent	102.09	2 mL	Colorless liquid; bp 139 °C; d 1.08 g/mL	
Sodium hypochlorite (bleach)	Oxidant → generates Br^+	74.44	18 mL	Pale greenish aqueous liquid	
Sodium bromide	Bromine source	102.89	~2.8 g	White crystalline solid	
Glacial acetic acid	Solvent / acid medium	60.05	10 mL	Colorless liquid; bp 118 °C; d 1.05 g/mL	
95% Ethanol	Rinse + recrystallization solvent	46.07	10 mL (+ 50% aq for recryst.)	Colorless liquid; bp 78 °C; d 0.79 g/mL	
Sodium thiosulfate	Quenches excess bromine	158.11	2 g	White solid	— low hazard
Sodium hydroxide	0.5 M work-up; 2 g solid to precipitate product	40.00	3 × 10 mL (0.5 M) + 2 g solid	Colorless soln / white pellets	

GRAB FIRST: 50-mL RBF + stir bar + clamp · **ice-water bath** + stir plate (plug into **outlet, not the Variac**) · 2× 1-mL green syringes + 25-mL Erlenmeyer + 400-mL beaker · **LATEX gloves** (aniline) then nitrile · steam bath · **125-mL Erlenmeyer** ×2 · Büchner funnel + filter flask + vacuum · grad cylinders nested in beakers w/ inverted-beaker caps · mortar & pestle (grind NaOH) · spatula.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Protect: aniline → N-phenylacetamide

- Clamp a **50-mL RBF + stir bar**. Instructor adds **2 mL acetic anhydride**. **Stopper the flask.**
- Lower the stoppered flask into an **ice-water bath** on a stir plate. **Bath needs water, not just ice, for good heat transfer. Plug the stir plate into an outlet, NOT the Variac.**
- Prep syringes: two **1-mL green syringes** in a 25-mL Erlenmeyer nested in a 400-mL beaker → carry to the aniline hood. **Put on LATEX gloves.**
- Draw **0.5 mL aniline into EACH** syringe (not 1 mL in one — the plunger pops out). **Record exact volumes** for theoretical yield.
- Add the aniline **dropwise over 5 min** with stirring. **The mixture will solidify as it reacts — break up the solid with a spatula so stirring continues.**
- Clean syringes at the aniline hood: draw **acetone** into each, empty into the **labeled aniline waste**; rinsed syringes + latex gloves → **red solid waste**. Switch to nitrile gloves.
- Add **10 mL 0.5 M NaOH** to the flask; warm on a **steam bath ~1 min** so solid stops sticking to the walls. Pour into a **125-mL Erlenmeyer**. Rinse the RBF with a **2nd 10 mL 0.5 M NaOH** into the same flask, then a **3rd 10 mL**.
- Cool to room temp, then **ice bath ~10 min**. Collect the colorless precipitate by **vacuum filtration**. Filtrate → aqueous waste.
- Reserve a small **IR-size sample** + take its IR. The rest goes into Part 2.

STOP: all aniline must be cleared from hoods/benches before Part 2 — **bleach must never contact aniline**. Instructor confirms.

Part 2 — Brominate

- Transfer the Part 1 solid into a **125-mL Erlenmeyer + stir bar**.
- Get **10 mL 95% ethanol** and **10 mL glacial acetic acid**. **Grad cylinder nested in a large beaker, capped w/ an inverted smaller beaker; keep the pipette vertical.**
- Add both to the flask; clamp and lower into an **ice bath** on the stir plate.
- Add **~2.8 g NaBr**; stir **5 min at 0 °C**.
- Measure **18 mL bleach**. **If the cylinder held acetone, rinse it with water first — never mix bleach + acetone.**
- Add the bleach to the reaction; stir **60 s**, then **remove the ice bath** and stir at room temp **15 min**.
- Re-cool in the ice bath**. Still stirring, add **2 g sodium thiosulfate** (quenches excess bromine). Then **grind 2 g NaOH** with a mortar & pestle and add it (fully precipitates the product).
- Stir **5 more min**. **Not everything will dissolve — the product is insoluble, so the mixture stays heterogeneous. That's expected.**
- Vacuum filter** the precipitate. Rinse the solid with **cold 50% aq. ethanol (2 × 5 mL)**, then keep pulling vacuum **2–3 min** to air dry. Filtrate → aqueous waste.

Part 3 — Recrystallize + analyze

- Add boiling stones to **~50 mL 50% aq. ethanol**; heat on a steam bath.
- Put the crude solid in a clean **125-mL Erlenmeyer**; recrystallize using the hot 50% aq. ethanol. **Add solvent a little at a time — too much and your product stays dissolved.**
- Collect crystals by **vacuum filtration**; air dry.

Experiment Notes (in lab)

Exact aniline volume: _____ mL

step (product)	mass	mp	% yield
Pt 1 — N-phenylacetamide	g	°C	
Pt 2/3 — 4-bromo product	g	°C	

Pt 1 — solidifying / color:

Pt 2 — on adding bleach (color/heat):

After thiosulfate + NaOH (ppt?):

Recrystallization (solvent used, crystals):

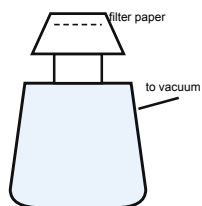
IR / para evidence:

22. **Weigh** the dry product → % yield; take **IR** (acetyl C=O, aromatic bands, para-substitution pattern).

✍️ **WRITE**

aniline is limiting → theoretical acetanilide → theoretical 4-bromo product; % yield for each step.

Setup — draw this



vacuum filtration (used 3×)

CONTEXT (don't copy) ▶

Why protect the amine? free aniline is so activated it over-brominates (tri-bromo) and its lone pair reacts with the oxidant. Acetylating dials it down to a controllable o/p-director → clean mono-**para**.

Where Br⁺ comes from: bleach (NaOCl) oxidizes Br⁻ from NaBr in acetic acid — greener than handling liquid Br₂.

Why thiosulfate then NaOH? thiosulfate destroys leftover bromine (safety + stops over-reaction); NaOH then drops the pH-sensitive product fully out of solution so you don't lose it in the filtrate.

Clean-up & Conclusions

- Aniline/acetone rinses → **labeled aniline waste**; latex gloves + syringes → red solid waste; ethanol-water filtrates → organic/aqueous waste as directed; keep bleach away from acetone & aniline; bench task.
- Conclusions** (~4 sentences [after lab](#)): % yield each step; that the amide directed bromination **para** (IR/NMR evidence); why protection was necessary; where product was lost (filtrate, over-solvation in recrystallization).